
SYNTHESIS CONDITIONS OF $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ AND STRUCTURALLY RELATED COMPOUNDS: EVIDENCE GRADING, ROUTE MATRIX, AND TRANSFERABILITY BOUNDARIES

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ABSTRACT

The non-centrosymmetric silicate $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ has been obtained by the high-temperature solution method and its single-crystal structure has been solved, but direct evidence for its starting composition, bulk phase region, and independent reproduction is still lacking. This review compares, with evidence grading, the published compositions, thermal histories, containers, cooling procedures, and characterization results for the target phase, the Zn-free tetragonal Ba–Y silicate family, structurally related Ba–Zn–Si–O compounds, and Y–Si–O process-reference systems. Existing experiments show that phase-formation behavior near the target is affected simultaneously by the Ba/Y/Si ratio, Zn–Y–oxygen stoichiometry, flux proportion, and cooling path; single-crystal structural evidence cannot substitute for bulk phase purity and actual composition analysis. On the basis of these results, the review proposes four priorities: establishing a local phase diagram around $\text{Ba}_5\text{Y}_{12}\text{ZnSi}_8\text{O}_{40}$, testing the composition–structure relationship of $\text{Ba}_5\text{Y}_{13-x}\text{Zn}_x[\text{SiO}_4]_8\text{O}_{8.5-x/2}$, comparing solid-state reaction and high-temperature solution crystal growth in parallel, and using Mg/Co analogues to distinguish size differences. A local two-step inorganic retrosynthesis model recommends ZnO, MgO, and Co_3O_4 , respectively, combined with BaCO_3 , Y_2O_3 , and SiO_2 as the preferred starting-material sets for the Zn, Mg, and Co targets; these results serve only as precursor hypotheses awaiting verification. The experiments most conducive to scientific discovery are not further trials of isolated temperatures but a closed experimental loop with feedback that links composition grids, phase quantification, single-crystal structure, actual stoichiometry, and process records.

1 INTRODUCTION

This review addresses the non-centrosymmetric barium–yttrium–zinc silicate $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$. It has three aims: to confirm the composition and structural assignment of the target phase; to clarify the phase-formation conditions under which the phase can be obtained reproducibly; and to determine whether a continuous composition–structure relationship exists between the Zn-containing phase and the Zn-free tetragonal Ba–Y silicates. The original report grew crystals by the high-temperature solution method in an open system and determined the structure by single-crystal X-ray diffraction (SCXRD) [1]. The structure combines a highly coordinated rare-earth–oxygen framework with isolated or composite silicate units, and it can serve as a reference for the structural assembly of complex silicates and for the compositional design of non-centrosymmetric oxides.

The direct report of the target phase establishes only the existence of the crystals and their structure; it does not provide an independently reproducible starting composition, a phase-formation range, or a preparation method for bulk phase-pure samples [1]. This review therefore does not assemble a “best recipe” from the conditions of structurally related compounds. Instead, it distinguishes the questions that single-crystal growth, solid-state reaction, and phase-diagram screening can each answer, and

uses this distinction to design experiments that discriminate between structural hypotheses. Literature values are retained throughout together with their stated limitations; structurally related systems and process-reference systems are used only to propose experimental references.

1.1 STATE OF RESEARCH ON RELATED SYSTEMS

The Ba–Y–Si–O tetragonal silicate family. Early flux-growth experiments reported $\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ as an accompanying crystalline phase; a subsequent set of 151 experiments in the $\text{BaO–K}_2\text{O–Y}_2\text{O}_3\text{–SiO}_2\text{–MoO}_3$ system showed that the $\text{Y}_2\text{O}_3/\text{SiO}_2$ ratio, the amount of MoO_3 , and the cooling rate jointly govern the structure type and crystal quality of the silicates [25, 42]. Modern single-crystal work revised the related nominal composition to non-stoichiometric $\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$; ceramic experiments further showed that grinding in agate may introduce SiO_2 and shift bulk samples away from the nominal ratio [43, 44]. A separate phase-region study combining energy-dispersive spectroscopy (EDS), continuous rotation electron diffraction (cRED), synchrotron powder X-ray diffraction (PXRD), and neutron diffraction confirmed $\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ and observed competing phases and a glass region [18]. These results indicate that the neighborhood of the target phase is a phase region controlled jointly by the Ba/Y/Si ratio, oxygen stoichiometry, liquid composition, and thermal history, rather than a single isolated recipe.

Structurally related compounds in Ba–Zn–Si–O. BaMSiO_4 ($M = \text{Co}, \text{Zn}, \text{Mg}$) belongs to the stuffed-tridymite-derived structure family, which shows that divalent tetrahedral cations can substitute for one another within a given framework, but this does not imply that these structures are isostructural with the Y-containing target phase [29]. The low-temperature and high-temperature structures of $\text{BaZn}_2\text{Si}_2\text{O}_7$ can be distinguished by neutron diffraction and X-ray diffraction, which shows that the observation temperature and the cooling path affect structure assignment [28]. A solid-state reaction study of $\text{BaZnSi}_3\text{O}_8$ proceeded through calcination, pelletizing, sintering, and controlled cooling in sequence, with characterization by synchrotron or laboratory X-ray diffraction [51]. The principal lesson of the Ba–Zn–Si–O literature is therefore to record the Zn source, atmosphere, cooling path, and phase composition at both high temperature and room temperature together, rather than to transfer any single temperature directly.

Y–Si–O process-reference systems. Both Y_2SiO_5 and $\text{Y}_2\text{Si}_2\text{O}_7$ exist in several polymorphs and have been prepared by the high-temperature solution method, melt growth, and the Czochralski method. Diffraction, local spectroscopy, and high-temperature characterization show that the individual polymorphs and the products obtained after cooling must be identified separately [6, 11, 31, 33]. These studies help standardize the recording of crucible, flux, volatilization loss, slow cooling, and annealing conditions, but they do not demonstrate that the same growth methods apply to $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$.

1.2 ORGANIZATION OF THIS REVIEW

This review first verifies the literature sources for the target phase and related structures, then summarizes what prior experiments have established about composition, competing phases, contamination, and thermal history, then compares the synthesis routes and their experimental conditions, and finally proposes four research directions: a local phase diagram, Zn–Y–oxygen-stoichiometry coupling, solid-state reaction in parallel with the high-temperature solution method, and Mg/Co analogues. The two-step inorganic retrosynthesis model used here generates only a set of precursors to be verified; temperatures, atmospheres, crucibles, and characterization methods are still determined from the literature and from materials-chemistry constraints.

Figure 1 summarizes the line of argument: related systems are first used to establish structural relationships, competing phases, and process boundaries; phase-diagram and charge-balance analysis proposes candidate compositions, from which the retrosynthesis model derives the precursor set; solid-state reaction and high-temperature solution crystal growth test the bulk phase region and the single-crystal structure, respectively; and composition and process are then adjusted round by round according to the phase composition, the actual stoichiometry, and the structure refinement results. Each experimental round should rule out at least one structural or phase-region hypothesis rather than pursue an unverified “best recipe”.

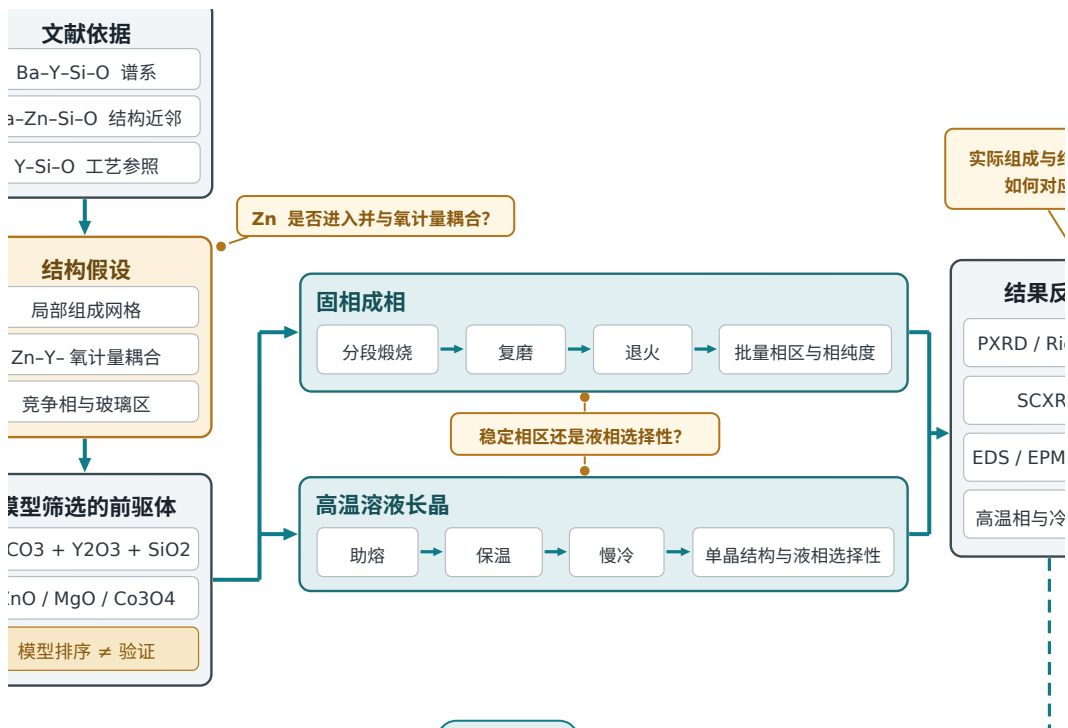


Figure 1: Research roadmap of this review. The Ba-Y-Si-O, Ba-Zn-Si-O, and Y-Si-O systems are first used to establish structural relationships, competing phases, and process boundaries; phase-diagram and charge-balance analysis proposes candidate compositions, and the two-step inorganic retrosynthesis model provides the precursor set; solid-state reaction and high-temperature solution crystal growth examine the bulk phase region and the single-crystal structure, respectively, and the characterization results are used to adjust subsequent compositions and processes.

2 LITERATURE SEARCH AND EVIDENCE EVALUATION

2.1 SEARCH SCOPE

The search started from the target chemical formula and its elemental form, using the spellings Ba₅Y₁₂Zn[O(SiO₄)]₈, Ba₅Y₁₂ZnSi₈O₄₀, BYZSO, and barium yttrium zinc silicate, combined with keywords such as space group, crystal structure, flux, solid-state, phase diagram, and crystal growth. Related literature was then traced along three directions: (i) the nomenclature and structural family of Zn-free tetragonal Ba-Y silicates; (ii) Ba-Zn-Si-O compounds with definite structural evidence; and (iii) Y-Si-O systems for which high-temperature solution, solid-state reaction, mechanical activation, or Czochralski processes are reported. The search first queried a local full-text materials library and then verified metadata and experimental passages through OpenAlex, Crossref, publisher pages, DOI resolution pages, IUCr, COD, RSC main texts and supplementary information (SI), and institutional repositories. The search cutoff date was 2026-09-01. A miss in a database or interface indicates only that the channel returned no result; it does not establish that the literature does not exist.

2.2 INCLUSION, EXCLUSION, AND DEPLICATION

A publication was included if it met any of the following conditions: it directly reports the preparation or structure of the target phase; it revises the structure, phase region, or compositional series of the tetragonal Ba-Y silicate family; or it provides traceable structural data or synthesis conditions for the structurally related Ba-Zn-Si-O or Y-Si-O systems. Review articles were used only to harmonize terminology and to trace original papers; specific experimental conditions were always taken from the original studies. The following sources were excluded: materials related only by element or function without a crystallographic or synthetic link; phases predicted computationally

but not realized experimentally; and aggregated abstracts or product information whose sources could not be traced. Glasses and glass-ceramics were treated as process-reference systems only where they clarify crystallization boundaries or competing phases; they do not support inferences about the structure of the target phase.

Records were first merged by DOI; where no DOI was available or several versions existed, records were cross-deduplicated by title, authors, year, and source, and author order, volume and pages, and publication status were checked. The search yielded 63 candidate records, of which 51 independent publications were retained after deduplication and source verification. Because only one direct original paper on the target phase exists, this review claims neither comprehensive coverage nor a completed independent reproduction.

2.3 LITERATURE RELEVANCE AND CONDITION EXTRACTION

Publications were divided into five categories according to the relationship between their subject and the target phase: direct reports of the target compound; Zn-free tetragonal Ba–Y silicates of the same family; structurally related compounds with a definite crystallographic relationship; synthesis studies usable as process references; and unrelated sources that cannot support structural or recipe judgments. Classification applies to individual claims rather than to whole papers; the structural results and process information from a single source can support interpretations of different scope.

To enable comparison, the following items were extracted from each publication: starting materials and nominal ratios, pretreatment, mode of phase formation, temperature–time, atmosphere and system openness, crucible or support, flux, cooling or growth program, and products and characterization methods. Numerical values were kept only within their corresponding original experimental records and were not merged across studies. Items not explicitly given in the original text were recorded as not reported, meaning that the information was not obtained in this check, not that the original paper definitely omitted it. Overall judgments gave priority to results on which several studies agree for the same experimental item, and single-crystal structure, bulk phase composition, actual stoichiometry, and high-temperature stability were examined separately.

3 THE TARGET COMPOUND AND RELATED STRUCTURES

3.1 QUESTION AND SCOPE

Before synthesis conditions can be compared, the object of study must be fixed. The available sources comprise the original report of the Zn-containing target phase, historical studies of Zn-free tetragonal Ba–Y silicates, and structurally related compounds that provide only structural references. The original report writes the target phase as $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ and determines its structure by single-crystal X-ray diffraction [1]; the latter two categories approach the target phase in composition or crystallographic description but cannot substitute for the Zn-containing target phase. This section verifies in turn the original chemical formula and its source and the revision of the historical nomenclature by modern structural work, and states the conclusions that the existing structural and phase-region evidence can support; the specific synthesis variables are given in Section 5.

This separates the question “which phase was obtained” from the question “which conditions were used”. For each record, the table retains the chemical formula as written in the original, the type-set form used here, the structural characterization result, and the open questions. Only studies in which both the chemical formula and direct structural evidence point to the target phase can serve as direct structural evidence for it. Sources on the same tetragonal Ba–Y silicate family are used to identify misassignment, non-stoichiometry, and competing phases; structurally related compounds only indicate local structural units that need checking and do not alter the definition of the target phase.

3.2 LITERATURE SOURCE OF THE TARGET PHASE

Among the publications checked in this review, $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ has only one direct report [1]. That report grew crystals by an open-system high-temperature solution method and identified the structure by single-crystal X-ray diffraction [1]. It is currently the only study that directly confirms the target phase, and its results cannot be combined with the conditions of related compounds into a

Table 1: Compositional and structural evidence for the target compound and related Ba–Y silicates.

Records are classified by the relationship between the subject of each publication and the target phase; neither Zn-free studies nor structurally related studies constitute an independent reproduction of the Zn-containing target phase.

Formula (original → typeset here)	Ref.	Main structural evidence	Relation to target phase	Conclusion and open questions
$\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$	[1]	Single-crystal X-ray structure determination [1]	Direct report of the target compound	Conclusion: Zn-containing target phase and its synthesis route type; open: independent reproduction, bulk phase-pure window, and correspondence to the Zn-free family
$\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ (tentative) (here: variable x retained)	[25, 42]	Accompanying-phase report and systematic phase-formation screening [25, 42]	Same-family tetragonal Ba–Y silicate	Conclusion: historical nomenclature and composition-screening results; open: exact stoichiometry, structural model, and relation to the target phase
$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ (here: non-stoichiometric formula retained)	[43]	Single-crystal revision as a non-stoichiometric modulated structure [43]	Same-family tetragonal Ba–Y silicate	Conclusion: modern structural interpretation of the historical formula; open: whether Zn can enter this family
$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ ceramics	[44]	Bulk phase composition and microstructure analysis [44]	Same-family tetragonal Ba–Y silicate	Conclusion: bulk characterization requirements for the non-stoichiometric family; open: distinguishing secondary phases and extraneous SiO_2 from the intrinsic phase region
$\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$	[18]	EDS, cRED, synchrotron PXRD, and neutron refinement [18]	Same-family tetragonal Ba–Y silicate	Conclusion: independently confirmed tetragonal Ba–Y structure and phase-region reference; open: site correspondence to the Zn-containing target phase
$\text{BaY}_{16}\text{Si}_4\text{O}_{33}$	[30]	Single-crystal structure with isolated orthosilicate units [30]	Structurally related compound	Conclusion: structural reference for Ba–Y orthosilicates; open: no evidence that it is the same phase as the Zn-containing target

“consensus recipe”. The report supports the target chemical formula, the route type, and the single-crystal structure assignment, but it does not demonstrate cross-laboratory reproduction or a bulk phase-pure window, nor does it fully resolve the oxygen-stoichiometry expression, the Zn site, or the relationship to the Zn-free tetragonal silicate family.

Table 1 lists the original spelling and the typeset form used here in separate columns. The latter only harmonizes symbols, subscripts, and brackets; it neither rewrites variable-composition formulas as fixed-composition ones nor modifies published chemical formulas on the basis of element conservation.

3.3 REVISION OF THE TETRAGONAL SILICATE FAMILY

Continued use of a historical name does not imply continuity of structure. An early flux-growth study recorded $\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ as a tentative accompanying phase [25]; a subsequent systematic flux

study included it in a multivariable phase-formation screening, but the subject remained the Zn-free tetragonal Ba–Y silicate [42]. Modern single-crystal work revised the related nominal composition to the non-stoichiometric modulated structure $\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ [43]; the corresponding ceramic study showed that bulk samples must also be checked for secondary phases and for SiO_2 introduced by the grinding medium [44]. Historical names must therefore be re-verified against structural models and bulk phase composition; none of these studies directly observed the exact Zn-containing target formula.

An independent phase-region study combining EDS, cRED, synchrotron PXRD, and neutron refinement confirmed tetragonal $\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ [18], providing a structural reference for Zn-free Ba–Y silicates that is distinct from the tentative nomenclature. The single-crystal structure of $\text{BaY}_{16}\text{Si}_4\text{O}_{33}$ gives a Ba–Y reference structure containing isolated orthosilicate units [30], which can be used to check the connectivity between polyhedra and silicate groups but is not direct evidence for the Zn-containing target phase. On this basis, this section fixes the chemical formula of the target phase; the remaining records serve to describe related structures and competing phases, and the scope of claims they can support is given in Section 4.

4 LITERATURE RELEVANCE AND COMPARABLE SCOPE

4.1 BASIS OF CLASSIFICATION

This review organizes the literature by the relationship between the subject of each study and the target phase, rather than assigning whole papers to a single category. A single study can provide evidence separately for structure, synthesis route, or characterization method, and each conclusion applies only to the object that the study directly observed. The literature is divided into five categories: direct reports of the target compound; Zn-free tetragonal Ba–Y silicates of the same family; structurally related compounds with a definite crystallographic relationship; synthesis studies usable as process references; and unrelated sources that cannot support judgments about the target structure or recipe. The classification describes the scope of applicability of each conclusion; it does not assess the quality of the papers.

Direct reports of the target compound. The chemical formula must match $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$, and the original study must give both the preparation procedure and the structure determination. At present only the open-system high-temperature solution growth and single-crystal X-ray structure study reported by Ababakeri et al. meets this condition [1]. That study provides the sole direct experimental starting point, but it has not established an independently reproducible bulk phase-pure window.

Same-family tetragonal Ba–Y silicates. These studies contain no Zn but bear directly on the historical nomenclature, structural revision, or phase-region relationships of tetragonal Ba–Y silicates. Early work reported the tentative accompanying phase $\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ [25]; modern single-crystal work revised the related composition to a non-stoichiometric modulated structure [43]; and an independent study confirmed tetragonal $\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ by several diffraction and compositional-analysis methods [18]. These results constrain the historical formulas, the composition range, and the competing phases, but they cannot substitute for a reproduction of the Zn-containing target phase.

Structurally related compounds, process references, and unrelated sources. Ba–Zn–Si–O or Ba–Y–Si–O compounds with reliable crystallographic data can be used to compare polyhedral connectivity, cation sites, and phase-transition characterization [28–30, 51]; studies that provide only synthesis route, atmosphere, cooling, or mechanical-treatment information can be used to design the experimental record items and to select characterization methods [31, 33, 41]. None of these sources allows the composition range of the target phase to be inferred. Studies that offer only elemental similarity, an application context, or general theoretical discussion, without a crystallographic or synthetic link, are not included in the condition comparison for the target phase [8, 12, 17].

Figure 2 shows the correspondence between the subject of each publication, the experimental conditions, and the questions that can be settled. Comparison first confirms which phase a publication actually studied and then compares starting materials, thermal history, atmosphere, container, and cooling conditions under the same experimental items. This arrangement facilitates comparison but does not remove the structural differences between compounds. The more distant the relationship

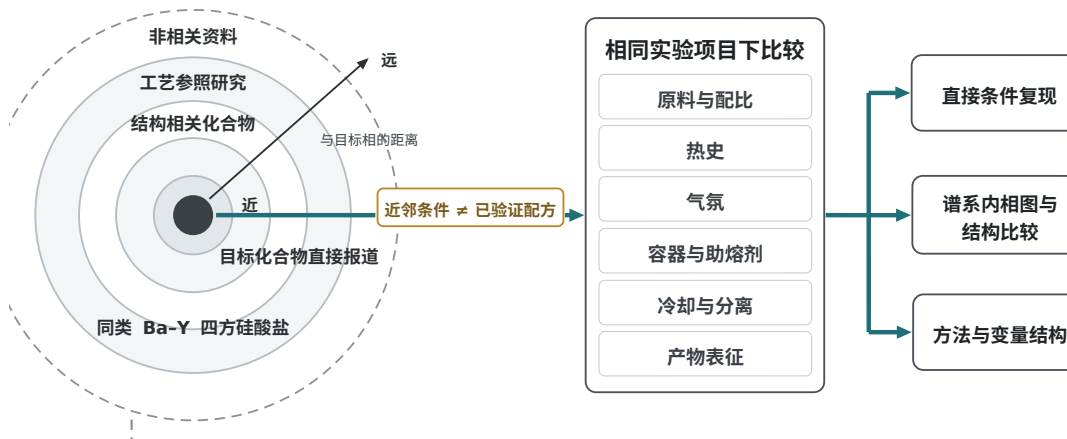


Figure 2: Scope of applicability of different literature categories to synthesis research on the target compound. The left column lists, in order, direct reports of the target compound, same-family tetragonal Ba–Y silicates, structurally related compounds, process-reference studies, and unrelated sources; the middle column organizes starting materials, thermal history, atmosphere, container, cooling, and product characterization under the same experimental items; the right column states the questions that each category can settle. At present there is only one direct report of the target compound [1]. Values from related systems serve only to illustrate experimental variables and characterization methods and cannot be taken directly as a verified recipe for the target phase.

between a publication and the target phase, the more the transferable content is confined to variable settings, experimental design, and characterization methods; specific temperatures, ratios, or flux amounts cannot be transferred directly.

4.2 STRUCTURAL RELATEDNESS

Structural relatedness requires at minimum traceable crystallographic evidence; it cannot rest on identical elemental composition alone. BaMSiO_4 ($M = \text{Co, Zn, Mg}$) has been assigned by single-crystal X-ray diffraction and powder neutron diffraction to the stuffed-tridymite-derived structure family [29]. This result can be used to compare the tetrahedral networks of Zn, Mg, and Co, but the family contains no Y and belongs to a different structural series, so it cannot serve as direct evidence for the target phase. The low-temperature and high-temperature structures of $\text{BaZn}_2\text{Si}_2\text{O}_7$ have been distinguished by neutron diffraction and X-ray diffraction [28]; the monoclinic structure of $\text{Ba}_2\text{ZnSi}_2\text{O}_7$ was established by single-crystal refinement [20]; and diffraction results for $\text{BaZnSi}_3\text{O}_8$ gave the connectivity network of ZnO_4 and SiO_4 [51]. These studies help define the structural parameters to be compared, but they cannot determine the conditions under which the target phase can be reproduced.

The complexity of four-coordinated Si networks in the Ba–Si–O system and their thermal response serve only as indirect structural references [16]. The polymorphs of $\text{Y}_2\text{Si}_2\text{O}_7$ and their thermal-expansion data show that one composition can correspond to different structures, so comparisons across decades must check the structural model and the observation temperature [5, 11]. These results can be used to frame structural characterization questions, but they do not provide the stability range of the target phase.

In summary, same-family Ba–Y silicate studies can be used to test composition and phase region, structurally related studies to compare sites and characterization methods, and process-reference studies to refine experimental design. Specific synthesis conditions can be interpreted only within the original object of study and its experimental range; they cannot be extrapolated directly because compounds are “similar”.

4.3 DELIMITING UNRELATED SOURCES

The presence of Ba and Si alone does not constitute structural relatedness. The structure refinement of benitoite-type BaSi_4O_9 [14], the trigonal structure of high-pressure BaSi_4O_9 [19], the structure refinement of tribarium silicate [40], and the structure report of high-temperature $\text{Ba}_2[\text{Si}_4\text{O}_{10}]$ [21] all show that Ba–Si–O networks can adopt different structures, but they lack the Y/Zn site correspondence required for the target phase. The report of 9R/6H perovskite-type structures in high-pressure Ba silicates further shows that a pressure route may lead to a different class of network, which cannot be used to infer the target route at ambient pressure [46].

Likewise, a similar application context cannot substitute for structural evidence. Optical studies of Ce^{3+} -doped $\text{Ba}_5\text{Si}_8\text{O}_{21}$ provide no information on the identity of the Y/Zn target phase [49]; non-centrosymmetric $\text{LiBa}_2\text{GaSi}_2\text{O}_8$ resembles the target phase only in its optical application context and differs in both composition and structural series [17]. The classic concept of stuffed silica structures and the theory–experiment review of zero-thermal-expansion materials help harmonize terminology and understand thermomechanical behavior, but they cannot support recipe judgments for the target phase [8, 12].

Unrelated sources are therefore used only to delimit the scope of the study and do not enter the comparison of synthesis conditions for the target phase. Even when related sources are listed in the same table, this indicates only that the experimental items are comparable, not that different compounds share the same structure or phase-formation conditions.

5 COMPARISON OF SYNTHESIS METHODS AND EXPERIMENTAL CONDITIONS

This section first summarizes the regularities obtained from earlier experiments and then compares the synthesis methods according to the main phase-formation processes. The tables list, item by item, starting materials and ratios, thermal history, atmosphere and system, container and flux, cooling or growth program, and product characterization. The purpose of the comparison is not to assemble a recommended recipe but to preserve the differences between the objects of study. The direct literature on the target phase specifies only an open-system high-temperature solution method and single-crystal X-ray structure identification; the charge, temperature program, crucible, flux and cooling program are not explicitly recorded and are therefore entered as not reported [1].

5.1 MAIN CONCLUSIONS OF EARLIER EXPERIMENTS

Existing experiments show that: first, composition coordinates explain the differences in phase formation in the Ba–Y–Si–O system better than a single maximum temperature does, and the $\text{Y}_2\text{O}_3/\text{SiO}_2$ ratio, the amount of MoO_3 and the cooling rate jointly affect the structure type and crystal quality [25, 42]; second, a single-crystal structure does not represent bulk phase purity, and Zn-free ceramic studies still detected secondary phases [18, 44]; third, a local phase diagram can convert a reproduction problem into a testable phase-region problem and provide a basis for the next round of charging [18]. On this basis, the following compares the experimental conditions of each route without directly rewriting the results of these related systems as a recipe for the target phase.

5.2 UNIFIED COMPARISON OF SYNTHESIS CONDITIONS

Figure 3 divides the literature records into three levels: “synthesis method–experimental conditions–product characterization”. The external-flux and self-flux/melt routes list flux proportion, container contact, cooling and separation, melt homogenization and nucleation conditions; the solid-state ceramic route lists mixing, pelletizing, calcination, regrinding and re-firing conditions; the mechanochemical route is included only when the literature explicitly reports mechanical activation; the Czochralski route additionally lists seed crystal, pulling, rotation and annealing conditions. The connecting lines in the figure indicate the correspondence between experimental conditions and characterization results and do not indicate that any given variable necessarily leads to a particular phase.

The number of publications for each method does not mean that their condition records are equally complete. Routes such as crucible-free melt growth, open-system high-temperature solution growth or high-temperature crystallization of rare-earth silicates can be confirmed only when the source

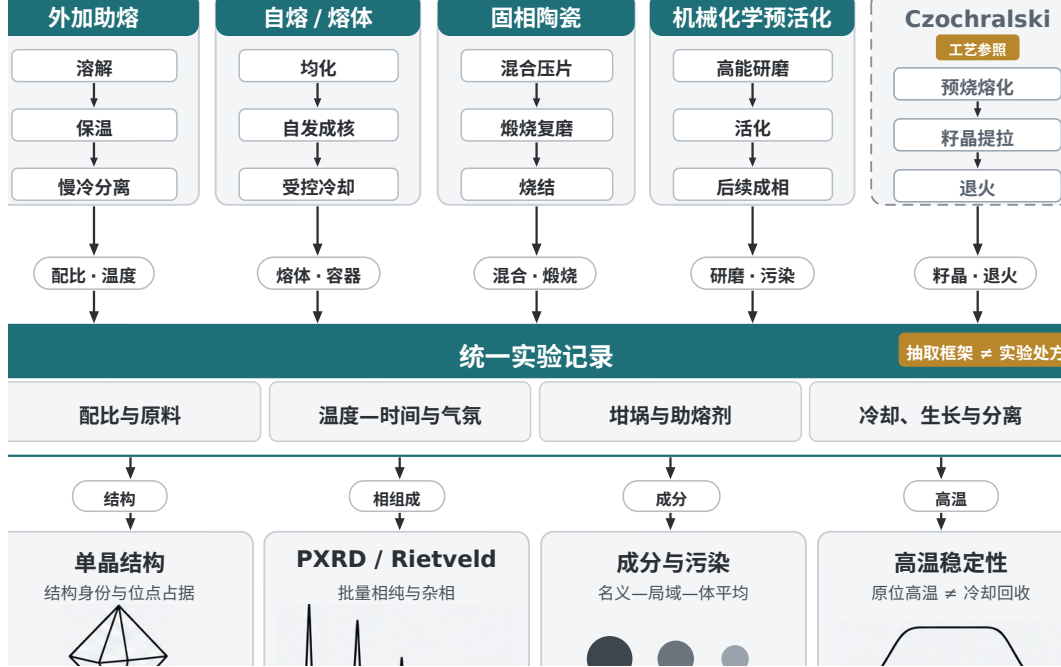


Figure 3: Comparison of conditions and product characterization across synthesis methods. The left-hand side distinguishes the external-flux, self-flux/melt, solid-state ceramic, mechanochemical pre-activation and Czochralski routes; the middle lists the comparable starting materials, thermal history, atmosphere, container, cooling and growth conditions; the right-hand side lists complementary characterization such as single-crystal structure, PXRD/Rietveld refinement, compositional analysis and high-temperature stability. The connecting lines indicate the correspondence between experimental conditions and characterization results, not a causal relation between variables and phase formation; the Czochralski route serves only as a process reference, and this figure does not constitute an experimental recipe for the target phase.

states them explicitly [2, 10, 15, 27]. Another title-level record by Leonyuk confirms only the single-crystal object and the structure refinement, and the growth method cannot be inferred from it [26]. The solid-state reaction literature covers structural phase transitions, ceramic sintering, dopant characterization and glass crystallization, but only items that can be located in an experimental passage are entered in the numerical tables [9, 22, 23, 28, 32, 35, 48, 51]. Among the mechanochemical sources, Tzvetkov and Minkova explicitly report mechanical treatment, whereas Yıldırım reports only additive-assisted solid–liquid-phase sintering, which cannot be classed as mechanochemical merely because the object of study is a powder [41, 45]. The Czochralski sources likewise serve only as a process reference for $\text{Y}_2\text{SiO}_5/\text{Ln}_2\text{SiO}_5$ [4, 7, 31, 47].

5.3 CONDITION SUMMARY TABLES

Table 2 lists, row by row, the literature source, chemical input, synthesis method and relation to the target phase of each record, together with the thermal history, system, crucible and flux, cooling or growth program, product characterization and items not recorded. Table 3 continues, under the same numbering, with the thermal history, atmosphere and system, crucible and flux, cooling or growth program, product characterization and items not recorded. When the source column is marked “experimental section”, this paper has checked the original experimental passage; when it is marked “abstract/title”, only the object or method explicitly stated there is confirmed, and steps not written out are not filled in from generic terms such as “crystal growth”. “—” indicates that no explicit record of the item was obtained in this check, not that the original experiment lacked the step.

Table 2: Summary of synthesis conditions for the target compound and related systems (I): literature source, chemical input, synthesis method and relation to the target phase. One row per record; zone A contains the direct literature on the target compound, zone B the Zn-free same-family tetragonal Ba–Y silicates (not reproductions of the target phase), and zone C structurally related and process-reference studies. “—” indicates that no explicit record was obtained in this check, not that the original experiment lacked the step; values for related systems are not validated recipes for the target phase.

No.	Source and location	Target/product formula	Starting materials, ratios and pre-treatment	Synthesis method; relation to target phase
Zone A: direct literature on the target compound				
A1	Publisher abstract; ESI used only for structure/composition verification [1]	$\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$	—	High-temperature solution; external flux/self-flux unclear; relation: direct report of the target compound
Zone B: Zn-free same-family tetragonal Ba–Y silicates, not reproductions of the target phase				
B1	p. 422, experimental section (accompanying-phase batch) [25]	$\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ (accompanying phase)	Whole-batch charge: $\text{BaCO}_3/\text{K}_2\text{CO}_3/\text{MoO}_3$ 1 g each, Y_2O_3 0.1614 g, SiO_2 0.1718 g; cannot be converted to a stoichiometric ratio for the accompanying phase	External flux; relation: same-family tetragonal Ba–Y silicate
B2	Abstract and systematic-experiment description; single-batch SI not obtained [42]	$\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ (one phase among 151 screening runs)	$\text{BaO}-\text{K}_2\text{O}-\text{Y}_2\text{O}_3-\text{SiO}_2-\text{MoO}_3$ study system; reagent form, single-batch ratio, pre-treatment —	External flux; relation: same-family tetragonal Ba–Y silicate
B3	Process excerpt from the original text (self-flux crystals) [43]	$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ ($x \approx 10.2$)	—	Self-flux/melt; relation: same-family tetragonal Ba–Y silicate
B4	Abstract-level evidence ($x = 9-11$ ceramic series) [43]	$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ (series)	—	Solid-state ceramic; relation: same-family tetragonal Ba–Y silicate
B5	p. 598, §2 [44]	$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ ($x = 9-14$ series)	$\text{BaCO}_3/\text{Y}_2\text{O}_3/\text{SiO}_2$; $\text{Ba}:\text{Y}:\text{Si}=x:26:16$ ($x = 9.0-14.0$); thermal pre-treatment of starting materials; dry milling in agate jar/balls at 400 rpm for 1 h, re-milled for 1 h after calcination	Solid-state ceramic; relation: same-family tetragonal Ba–Y silicate
B6	Main text §2.1.1, SI §1.1/§1.3 [18]	$\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ (phase A)	$\text{BaCO}_3/\text{Y}_2\text{O}_3/\text{SiO}_2$; $\text{BaCO}_3:\text{YO}_{1.5}:\text{SiO}_2 = 5:13:8$, final charge 0.5 g; drying/pre-heating, agate grinding, 10 mm pellets	Solid-state ceramic/powder; relation: same-family tetragonal Ba–Y silicate
Zone C: structurally related and process-reference high-temperature solution and melt studies				
C1	Publisher abstract [2]	$\text{Ba}_3\text{Zn}_4\text{Si}_4\text{O}_{15}$	—	High-temperature solution; external/self-flux unclear; relation: structurally related compound
C2	Experimental section of the IUCr original [33]	$\beta\text{-Y}_2\text{Si}_2\text{O}_7$, with $\text{Na}_2\text{Si}_2\text{O}_5$ and oxyapatite	$\text{Na}_2\text{CO}_3/\text{Y}_2\text{O}_3/\text{SiO}_2$ in NaYSi_2O_6 proportions; nutrient: $\text{Na}_2\text{MoO}_4 = 1:10$; pre-treatment —	External flux; relation: process-reference study
C3	Record from the original paper [15]	Zn_2SiO_4	Melt $5\text{Zn}_2\text{SiO}_4 + 3\text{Pb}_2\text{ZnSi}_2\text{O}_7$; trials with fluoride as Zn source; pre-treatment —	External flux; relation: process-reference study
C4	Title/metadata [10]	High-temperature rare-earth pyrosilicates	—	Crucible-free melt; relation: process-reference study

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Table 2: Summary of synthesis conditions (I) (continued)

No.	Source and location	Target/product formula	Starting materials, ratios and pre-treatment	Synthesis method; relation to target phase
C5	Indexed experimental passage of the original (flux branch) [27]	$\text{Y}_2\text{SiO}_5/\text{Y}_2\text{Si}_2\text{O}_7/\text{LaBSiO}_5$	Y–Si–O nutrient; Li/K di- and trimolybdates; exact ratio, pre-treatment —	External flux; relation: process-reference study
C6	Title and DOI metadata [26]	$\text{Y}_2\text{SiO}_5/\text{Y}_2\text{Si}_2\text{O}_7/\text{LaBSiO}_5$	—	Unclear (title does not distinguish the growth route); relation: process-reference study
Zone C: structurally related and process-reference solid-state ceramic studies, first group				
C7	Experimental section of the original [28]	$\text{BaZn}_2\text{Si}_2\text{O}_7$	$\text{BaCO}_3/\text{ZnO}/\text{SiO}_2$ in stoichiometric ratio; thorough mixing with intermediate grinding	Solid-state ceramic; relation: structurally related compound
C8	Experimental section of the original paper PDF [51]	$\text{BaZnSi}_3\text{O}_8$	BaCO_3 (99.8%), ZnO/SiO_2 (99.5%); ratio —; ball milling in polyethylene jar/ ZrO_2 balls/deionized water for 5 h, dried at 85 °C, pelletized at 150 MPa	Solid-state ceramic; relation: structurally related compound
C9	Experimental section of the original paper; batch mapping not exposed [23]	$\text{Ba}_2\text{ZnSi}_2\text{O}_7/\text{BaZnSiO}_4$; $\text{BaZn}_{2-x}\text{Mg}_x\text{Si}_2\text{O}_7$ series	$\text{SiO}_2/\text{BaCO}_3/\text{ZnO}$; Mg source is a carbonate–hydroxide hydrate; ball milling; single-batch ratio/milling parameters —	Solid-state ceramic; relation: process-reference study
C10	Title/metadata [9]	$\text{BaZnSi}_3\text{O}_8$ -based ceramics	—	Solid-state ceramic; relation: process-reference study
C11	Title/metadata [48]	$\text{Ba}_2\text{MgSi}_2\text{O}_7$ ceramics	—	Solid-state ceramic; relation: process-reference study
Zone C: structurally related and process-reference solid-state ceramic studies, second group				
C12	Experimental section of the open-access original [35]	$\text{Ba}_2\text{ZnSi}_2\text{O}_7:\text{Sm}^{3+}$	$\text{BaCO}_3/\text{ZnO}/\text{SiO}_2$ (99%), Sm_2O_3 (99.9%); nominal doping ratio —; grinding in agate mortar under ethanol for 1 h	Solid-state ceramic; relation: process-reference study
C13	Original abstract [32]	Ba/Zn orthosilicates: $\text{Eu}^{3+}/\text{Mn}^{2+}$	Oxides/carbonates, or $\text{Ba}_2\text{SiO}_4:\text{Eu}$ and $\text{Zn}_2\text{SiO}_4:\text{Mn}$ precursors; ratio, pre-treatment —	Solid-state reaction; relation: process-reference study
C14	Thesis metadata/process index [22]	Ba–Zn–Si crystalline phases/crystallized phases of high-temperature glass solders	—	Solid-state ceramic (glass-crystallization context); relation: process-reference study
C15	Institutional repository abstract [45]	Y_2SiO_5 powder	LiYO_2 additive; ratio, pre-treatment —	Solid–liquid-phase sintering; not evidence of mechanochemistry; relation: process-reference study
C16	Indexed experimental section of the original [50]	$\text{Ba}_2\text{ZnSi}_2\text{O}_7$	BaCO_3 (99.8%), ZnO/SiO_2 (99.5%); ratio —; ball milling in polyethylene jar/ ZrO_2 balls/deionized water for 5 h	Solid-state ceramic; relation: process-reference study

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Table 2: Summary of synthesis conditions (I) (continued)

No.	Source and location	Target/product formula	Starting materials, ratios and pre-treatment	Synthesis method; relation to target phase
Zone C: mechanical activation and Czochralski process references				
C17	p. 126, §2 [41]	$\text{Y}_2\text{SiO}_5/\text{Y}_2\text{Si}_2\text{O}_7/$ yttrium oxyapatite mixed phases	$\text{Y}_2\text{O}_3/\text{SiO}_2 = 7:9$; hydrated yttria also compared; 80 cm ³ agate jar, 20 mm/total 40 g agate balls, ball-to-powder ratio 7:1; in air at 256 rpm, 1–3 h	Mechanochemical pre-activation + subsequent heat treatment; relation: process-reference study
C18	Experimental section of the original [31]	Y_2SiO_5	Y_2O_3 (99.999%)/ SiO_2 (99.99%), about 980 g, stoichiometric ratio; mixed and ground in corundum mortar, hydraulically pressed into blocks and pre-fired	Czochralski; relation: process-reference study
C19	Indexed experimental passage of the original (Czochralski branch) [27]	Y_2SiO_5 (including Cr-doped series)	High-purity $\text{Y}_2\text{O}_3/\text{SiO}_2$; exact ratio, pre-firing —	Czochralski; relation: process-reference study
C20	Index of the growth chapter of the thesis [4]	Rare-earth-doped Y_2SiO_5	—	Czochralski (chapter-level index); relation: process-reference study
C21	Indexed experimental passage of the original [47]	$\text{Y}_2\text{SiO}_5/\text{Eu}:\text{Y}_2\text{SiO}_5$	Starting materials of purity $\geq 99.99\%$; to counter SiO_2 volatilization, the melt is offset from the stoichiometric composition by at most 0.2% (direction unclear as worded in the original); pre-firing —	Czochralski; relation: process-reference study
C22	Abstract of the original route study [7]	Ln_2SiO_5 (including Y_2SiO_5)	—	Czochralski; relation: process-reference study

Table 3: Summary of synthesis conditions (II): thermal history, atmosphere and system, crucible and flux, product characterization and items not recorded. Numbering corresponds to Table 2; the labelled sub-items separated by “;” within a cell (cooling/growth, crucible/flux) carry over the corresponding columns of the original table.

No.	Temperature–time; cooling/growth	Atmosphere/ system; crucible/flux	Products and characterization	Items not recorded
Zone A: direct literature on the target compound				
A1	—	Open system; gas composition —	Single crystals; SCXRD; yield, size and bulk phase purity —	Starting materials, ratio, pre-treatment, temperature–time, gas, crucible, flux, cooling/ separation, yield/size, phase fraction

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Table 3: Summary of synthesis conditions (II) (continued)

No.	Temperature–time; cooling/growth	Atmosphere/ system; crucible/flux	Products and characterization	Items not recorded
Zone B: Zn-free same-family tetragonal Ba–Y silicates, not reproductions of the target phase				
B1	1150 °C, 3 h; cooling/growth: cooled at 2 K h ⁻¹ to 900 °C; slow cooling to room temperature after switching off the furnace	Air; open/closed nature unclear; crucible lidded; crucible/flux: lidded Pt crucible; MoO ₃ 1 g; flux phase dissolved in distilled water	Colourless prismatic accompanying crystals; single-crystal refinement gives a tentative formula; separate yield, bulk phase fraction —	Separate stoichiometric ratio/ pre-treatment of the accompanying phase, system closure, separate yield, phase fraction
B2	—	—	Multiphase screening; SCXRD/PXRD/Rietveld/ SEM; single-batch yield of this phase —	Single-batch ratio, pre-treatment, temperature–time, atmosphere, crucible, flux amount, cooling, yield
B3	1600 °C; time —	—	Self-flux single crystals; SCXRD; yield, size, bulk phase fraction —	Starting materials, ratio, pre-treatment, time, system, crucible, flux, cooling, yield/ size, phase fraction
B4	—	—	Ceramic composition window; specific roles of SCXRD/PXRD and phase fraction —	Starting materials, single-point ratio, pre-treatment, entire thermal history/system/ container/cooling, phase fraction
B5	Calcination at 1300 °C, 12 h; firing at 1600 °C, 2 h; cooling/growth: furnace-cooled to room temperature after power-off	Air; system closure —; crucible/flux: Pt–Rh plate; lidded alumina crucible; no external flux listed	Ceramics; PXRD; secondary phases and SiO ₂ contamination from agate observed; quantitative phase fraction —	Heating/cooling rates, system closure, yield/phase fraction
B6	Calcination at 1273 K, 12 h; annealing at 1573 K, 24 h with regrinding and re-firing twice; neutron sample additionally homogenized at 1573 K for 12 h; cooling/growth: crushed and finely ground after cooling; not crystal growth	Air; heating/cooling at 5 K min ⁻¹ ; crucible/flux: alumina crucible; no flux listed	Powder; PXRD monitoring; EDS, cRED, synchrotron PXRD, neutron TOF; yield/ quantitative phase fraction —	Separate yield, quantitative phase fraction
Zone C: structurally related and process-reference high-temperature solution and melt studies				
C1	—	Open to air; gas flow —	Single crystals; SCXRD; yield, size, phase fraction —	Starting materials, ratio, pre-treatment, temperature–time, gas flow, crucible/flux, cooling/ separation, yield/size/phase fraction
C2	Slowly heated to 1473 K (1200 °C), 24 h; cooling/growth: cooled at 2 K h ⁻¹ to 673 K (400 °C); subsequent cooling —	Atmosphere, open/ closed —; crucible lidded; crucible/flux: lidded Pt crucible; Na ₂ MoO ₄ ; flux dissolved in boiling water	Colourless cube-shaped crystals and accompanying phases; SCXRD at 100/ 280 K; yield —	Pre-treatment, atmosphere/ system, subsequent cooling, yield

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Table 3: Summary of synthesis conditions (II) (continued)

No.	Temperature–time; cooling/growth	Atmosphere/ system; crucible/flux	Products and characterization	Items not recorded
C3	1300 °C to 960 °C; dwell —; cooling/growth: 1 °C h ⁻¹ ; subsequent cooling —	Crucible/flux: crucible —; Pb ₂ ZnSi ₂ O ₇ / fluoride flux; separation —	Hexagonal prisms of several millimeters; chemical analysis/buoyancy density, source formula Zn _{1.96} Si _{1.04} O _{4.04} ; yield —	Pre-treatment, dwell, system, crucible, separation, subsequent cooling, yield
C4	—	Crucible/flux: crucible-free; flux —	Melt product; neutron powder diffraction/profile refinement; yield, phase fraction —	Starting materials, ratio, pre-treatment, temperature–time, system, flux, cooling, yield/phase fraction
C5	—	Crucible/flux: Li/K molybdate flux; crucible, flux ratio, separation —	Spontaneously nucleated crystals; electron microprobe, XRD/structure refinement; yield —	Exact ratio, pre-treatment, temperature–time, system, crucible, flux ratio/separation, cooling, yield
C6	—	—	Title supports only the single-crystal object and structure refinement; yield/ size —	Starting materials, ratio, pre-treatment, synthesis method and all experimental conditions, yield/size
Zone C: structurally related and process-reference solid-state ceramic studies, first group				
C7	1280 °C; time —	—	White polycrystalline powder; PXRD, neutron diffraction, GSAS/Rietveld; yield/phase fraction —	Time, system, crucible/flux, cooling, yield/phase fraction
C8	Calcination in air at 1000 °C, 3 h (5 °C min ⁻¹); sintering at 1090–1120 °C, 3 h; cooling/growth: cooled at 2 °C min ⁻¹ to 1000 °C, then furnace-cooled	Air; open/closed —	Ceramics; laboratory/ synchrotron XRD with Rietveld; yield/quantitative phase fraction —	Ratio, system closure, crucible/ flux, yield/phase fraction
C9	Cooling/growth: ball-milled again after cooling; cooling rate —	—	Polycrystalline; HT-XRD, dilatometer; single-batch phase fraction —	Single-batch ratio/milling, temperature–time, system, crucible/flux, cooling rate, phase fraction
C10	—	—	Ceramics; structural/ microwave dielectric characterization; specific phase-purity method and phase fraction —	Starting materials, ratio, pre-treatment and all experimental conditions, phase fraction
C11	—	—	Ceramics; structural/ sintering/microwave dielectric characterization; specific phase-purity method and phase fraction —	Starting materials, ratio, pre-treatment and all experimental conditions, phase fraction
Zone C: structurally related and process-reference solid-state ceramic studies, second group				
C12	In O ₂ at 1200 °C, 6 h; heating at 5 °C min ⁻¹ ; cooling/growth: natural cooling to room temperature, then grinding	O ₂ ; open/closed —; crucible/flux: alumina crucible; flux —	Powder; XRD/FTIR/ SEM–EDS/thermal and spectroscopic; yield/phase fraction —	Doping ratio, system closure, flux, yield/phase fraction
C13	—	—	Powder; PXRD/IR/UV–vis/ luminescence; yield/phase fraction —	Ratio, pre-treatment and all experimental conditions, yield/ phase fraction

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Table 3: Summary of synthesis conditions (II) (continued)

No.	Temperature–time; cooling/growth	Atmosphere/ system; crucible/flux	Products and characterization	Items not recorded
C14	—	—	Crystallised phases; HT-XRD/dilatometry characterization index; single-batch phase fraction —	Starting materials, ratio, pre-treatment and all single-batch experimental conditions, phase fraction
C15	—	Crucible/flux: crucible —; LiYO ₂ additive amount —	Powder; optical microscopy, XRD, SEM–EDS; yield/ phase fraction —	Ratio, pre-treatment and entire thermal history/system/ container/cooling, yield/phase fraction
C16	Calcination in air at 1100 °C, 3 h; sintering in air/O ₂ /N ₂ at 1200 °C or in N ₂ –1 vol% H ₂ at 1125 °C; sintering time —	air/O ₂ /N ₂ / N ₂ –1 vol% H ₂ ; open/closed —	Ceramics; XRD/XPS/ microwave dielectric; BaSiO ₃ observed in reducing-atmosphere samples; yield/phase fraction —	Ratio, sintering time, system closure, crucible/flux, cooling, yield/phase fraction
Zone C: mechanical activation and Czochralski process references				
C17	Subsequent heat treatment in air for 2 h; the abstract summarizes phase formation as $T > 1000$ °C, and the specific set points are not fully exposed by this passage	Milling and heat treatment both in air; open/closed nature —; crucible/flux: agate balls/jar used for milling; crucible and flux for the synthesis heat treatment —	Mixed-phase powder; XRD/ IR/thermal analysis; phase fraction —	Specific temperature points of the synthesis heat treatment, system closure, heat-treatment crucible/flux, cooling, phase fraction
C18	Single-phase charge prepared at 1200 °C, 24 h; melting temperature/growth time —	High-purity N ₂ ; open/closed —; crucible/flux: Ir crucible, 80×60 mm; flux not applicable	Single crystal; orientation, optical microscopy, SEM, absorption spectra; structural phase purity/yield —	Melting temperature/growth time, system closure, pulling/ rotation rate/cooling, structural phase purity/yield
C19	Cooling/growth: pulling at 2.0–2.5 mm h ^{–1} ; rotation, cooling/ annealing —	Crucible/flux: RF-heated Ir crucible, about 80×90 mm; flux not applicable	Large crystals/spontaneously nucleated crystals; electron microprobe, XRD/structure refinement; yield —	Exact ratio/pre-firing, melting/ time, system, rotation/cooling/ annealing, yield
C20	—	Crucible/flux: crucible —; flux not applicable	Crystals/spectroscopic and crystal-field analysis; structural phase purity, yield/ size —	Starting materials, ratio, pre-firing and all growth parameters, structural phase purity/yield/size
C21	Melting temperature, growth time —; post-growth annealing for 20 h	Crucible/flux: RF-heated Ir crucible; flux not applicable	Crystals of about 35 mm×130 mm, 460–486 g; Ir inclusions observed microscopically; structural phase purity/yield —	Pre-firing, melting/growth time, system, pulling/rotation/ cooling, annealing temperature, structural phase purity/yield
C22	—	Crucible/flux: crucible —; flux not applicable	Single crystals; melt stability, dopant partitioning and facet formation studied; structural phase purity/yield —	Starting materials, ratio, pre-firing and all growth parameters, structural phase purity/yield

5.4 COMPARISON BY ZONE

The core message of zone A is that the direct literature does not give a complete recipe: for the target compound, only the route category and the SCXRD result are currently available for checking, and no recipe item can be back-filled from related systems [1]. Zone B shows that studies of Zn-free same-family Ba–Y silicates already cover flux-grown accompanying phases, self-flux crystals, ceramic composition scans and multi-probe solid-state/powder phase-region studies, but none of the objects

of study is the Zn-containing target phase; in particular, the systematic screening for which single-batch SI was not obtained cannot be compressed into a single “representative condition” [18, 25, 42–44].

The direct value of zone C lies in showing the differences in experimental conditions between methods. High-temperature solution literature usually gives the flux composition, container and slow-cooling/separation conditions together, whereas solid-state reaction literature more often gives mixing, calcination, regrinding and sintering conditions; when the same system is prepared under different atmospheres, volatilization loss and secondary phases must also be treated as product information, and the atmosphere cannot be regarded as an isolated “optimization variable” [33, 50, 51]. C17 gives the rotation rate, duration, agate balls/jar and ball-to-powder ratio, so its mechanical activation step can be described, but the product remains a mixed-phase powder and cannot supply a recipe for the target phase; Czochralski studies, even when they give the charge, crucible and crystal size, still require structural and phase-purity characterization, and the growability of the target phase cannot be inferred from crystal appearance [27, 31, 41, 47]. Therefore, the values in the tables hold only within their respective literature records: they are not merged across publications, not averaged over different compounds, and “not given in the abstract” is not rewritten as “not reported in the original”.

6 INTEGRATION OF PRIOR EXPERIMENTAL FINDINGS

Comparing conditions not only preserves the numerical values but also reveals patterns that recur across different studies. Integrating the published experiments on the target phase and its related compounds yields three insights that can directly guide the next round of experiments.

6.1 THE DOMINANT FACTOR IN PHASE-FORMATION DIFFERENCES: COMPOSITION COORDINATES RATHER THAN A SINGLE MAXIMUM TEMPERATURE

The 151 flux-growth experiments in the Ba–Y–Si–O system show that the $\text{Y}_2\text{O}_3/\text{SiO}_2$ ratio and the amount of MoO_3 govern the connectivity of the silicate groups and the type of crystal obtained. Lowering the $\text{Y}_2\text{O}_3/\text{SiO}_2$ ratio, or increasing the MoO_3 content within a certain range, shifts the structures toward higher SiO_4 connectivity; an excess of MoO_3 is detrimental to crystal growth, and lower cooling rates generally improve crystal morphology and yield [42]. Screening for the target phase should therefore treat the nutrient composition, the flux ratio, and the cooling rate as independent variables rather than merely replicating a particular maximum temperature. The early accompanying crystals were obtained after soaking at 1150 °C followed by slow cooling, but that result came from a whole-batch flux system containing $\text{K}_2\text{CO}_3/\text{MoO}_3$ and cannot be converted into a stoichiometric starting composition for the target phase [25].

Solid-state studies point to the same conclusion. In $\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ ceramics across the $x = 9\text{--}14$ composition series, the main phase and the secondary phases vary with the nominal Ba content, and SiO_2 introduced by agate grinding interferes with the identification of the intrinsic phase region [44]. $\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ was located through a powder-composition screening that traversed the phase diagram and was confirmed by repeated annealing and grinding together with multiple diffraction and compositional-analysis methods [18]. What the target phase currently lacks is therefore not yet another isolated temperature but a local phase diagram of the Ba–Y–Zn–Si–O system, together with quantitative phase analysis for each composition coordinate.

6.2 DISTINGUISHING SINGLE-CRYSTAL EVIDENCE FROM THE BULK PHASE-PURITY WINDOW

The direct report on the target phase confirmed its structure by single-crystal diffraction, but the available material does not also provide an independently reproducible starting composition, thermal history, or bulk phase fraction [1]. Ceramic studies of the Zn-free series show that even once the main phase has formed, bulk samples may still contain secondary phases and contamination introduced by the processing media [18, 44]. A reproduction experiment therefore requires at least two complementary types of characterization: confirmation of the structural model by single-crystal or advanced diffraction methods, and determination of the target-phase fraction in the whole batch by quantitative PXRD/Rietveld analysis. Without either one, “obtaining target crystals” cannot be equated with “establishing a phase-pure preparation method”.

6.3 THE LOCAL PHASE DIAGRAM: THE CENTRAL EXPERIMENT LINKING REPRODUCTION AND DISCOVERY

The discovery of $\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ shows that even when many compounds are already known in the $\text{BaO}\text{--}\text{Y}_2\text{O}_3\text{--}\text{SiO}_2$ phase region, a new phase may still lie at the boundary of competing phases or of a glass-forming region. The investigators narrowed the target region only gradually, updating the next batch of starting compositions round by round on the basis of the overall composition, the known phase proportions, and compositional analysis [18]. The Zn-containing system needs this strategy even more, because the introduction of Zn simultaneously changes the cation count, the oxygen stoichiometry, and possibly the liquid-phase behavior. Phase-diagram experiments can distinguish three situations: a line compound of narrow composition, a limited solid-solution range within the Zn-free series, and a metastable crystalline phase that forms only along a specific liquid-phase path. Compared with repeated trials at a single temperature, each of these three outcomes provides more structural information.

7 SYNTHESIS METHODS OF RELATED COMPOUNDS

This section compares the synthesis methods of related compounds according to the phase-formation process rather than listing the literature paper by paper. Figure 3 summarizes the comparable experimental items, and Table 2 retains the numerical values and literature sources. The discussion below is confined to the methodological information provided by the related studies; no route is rewritten as a candidate recipe for $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$.

7.1 HIGH-TEMPERATURE SOLUTION ROUTES

The literature on the Y–Si–O system first shows that “obtaining crystals” is not the same as “assigning the structure correctly”. The $\beta\text{-Y}_2\text{Si}_2\text{O}_7$ obtained with Na_2MoO_4 as the flux was accompanied by other silicate crystals, and the final phase identity was confirmed by single-crystal structure determination; such studies provide reference points for the nutrient-to-flux ratio, crucible lidding, slow cooling, and crystal separation, but they cannot give the stability range of the target phase [33]. A re-examination of the $\text{Y}_2\text{Si}_2\text{O}_7/\text{Y}_2\text{SiO}_5$ polymorphs by ^{89}Y MAS-NMR shows that local probes can complement diffraction methods in distinguishing polymorphs but cannot by themselves prove the phase formation of the target phase [6]. The review of Y–Si–O synthesis and the classic crystal-chemical classification of rare-earth silicates can unify the terminology for polymorphs and silicate polymerization degree, but they cannot replace the original experimental data [13, 34].

In the Ba–Zn–Si–O system, the open-system high-temperature solution study confirmed only the structure of $\text{Ba}_3\text{Zn}_4\text{Si}_4\text{O}_{15}$ and the category of the route; the exact batch and thermal history remain recorded as not reported [2]. The early molten-solution growth study of Zn_2SiO_4 shows that the slow-cooling procedure, the crystal-separation method, and the flux composition must be given together; conditions involving Pb/F additionally require review of toxicity, volatilization, and waste disposal, and cannot serve as a default choice [15]. The crucible-free melt study with a clearly traceable source and the high-temperature crystallization study of $\text{Y}_2\text{SiO}_5/\text{Y}_2\text{Si}_2\text{O}_7$ provide reference points for container contact, spontaneous nucleation, and structural identification, but they cannot prove that the target phase exhibits the same melt-growth behavior [10, 27]. The other paper by Leonyuk offers only title-level evidence; it confirms only its single-crystal subject and structure refinement, and no conclusion about the synthesis method can be drawn from it [26].

Implications for experiments on the target phase. The starting composition and the flux composition should be listed separately, the container contact should be noted, and the cooling and separation procedures should be recorded in full; crystal-structure determination, bulk diffraction, and compositional analysis should be carried out as a matched set. None of the following can be extrapolated directly: the specific values for related compounds, the applicability of Pb/F systems, or the assignment of the target-phase identity from crystal habit.

7.2 SOLID-STATE REACTION ROUTES

For BaMSiO_4 ($M = \text{Co}, \text{Zn}, \text{Mg}$), single-crystal X-ray diffraction and powder neutron diffraction established an isostructural stuffed-tridymite series, showing that isovalent substitution on the tetra-

hedral site can preserve the parent topology; this framework, however, cannot be equated with the zero-dimensional mixed building units of the target phase [29]. The single-crystal refinement of monoclinic $\text{Ba}_2\text{ZnSi}_2\text{O}_7$ and the structure report on monoclinic $\text{Ba}_2\text{MgSi}_2\text{O}_7$ provide the Zn and Mg references, respectively; both can be used to identify competing phases and to check substitution risks, but neither supports the assertion that they are isostructural with the target phase [3, 20].

Controlled substitution series demonstrate how composition, structure, thermal history, and characterization should be reported as a matched set. The Ba/Sr-substituted compounds were confirmed jointly by SCXRD and EDS; the HT-XRD results for $\text{BaZn}_{2-x}\text{Co}_x\text{Si}_2\text{O}_7$ and for the Ba/Sr–Zn–Si/Ge series gave composition-dependent regions of structural retention and phase boundaries. These studies show that the nominal degree of substitution must be matched to the measured composition and the in situ structural results, and that the Y/Zn tolerance of the target phase cannot be quantified across systems [24, 36, 38]. The review of $\text{BaZn}_2\text{Si}_2\text{O}_7$ solid solutions can organize the background on how thermal expansion varies with composition, but it cannot replace an individual original sintering batch [37]. The glass-ceramic study containing $\text{Ba}_{1-x}\text{Sr}_x\text{Zn}_2\text{Si}_2\text{O}_7$ likewise provides only functional and crystallization background and cannot serve as a direct basis for synthesizing the target phase [39]. The controlled comparison of $\text{Ba}_2\text{ZnSi}_2\text{O}_7$ ceramics under multiple atmospheres shows that atmosphere changes must be interpreted together with Zn loss, secondary phases, and surface valence states; structural stability cannot be judged from changes in dielectric properties alone [50].

Implications for experiments on the target phase. The stoichiometric weighing, regrinding and refring, atmosphere, and measured composition should be given side by side, with characterization by PXRD/Rietveld analysis, compositional analysis, and HT-XRD where necessary. The isovalent-substitution, thermal-expansion, or dielectric results of related studies cannot be rewritten directly as the aliovalent-substitution mechanism, phase-pure range, or reduction stability of the target phase.

7.3 MECHANOCHEMICAL PRE-ACTIVATION AND CZOCHRALSKI PROCESS REFERENCES

Mechanochemical studies support only the insight that “pre-activation may alter the subsequent phase-formation path”. The yttrium oxyapatite study gave mechanical-treatment conditions of 256 rpm in air for 1–3 h with an agate jar and agate balls; heat treatment was still required afterwards, and several Y–Si–O phases coexisted. These values describe only that mixed-phase system and cannot be applied directly as activation conditions for the target phase [41]. The Y_2SiO_5 powder study explicitly reported only LiYO_2 -assisted solid–liquid-phase sintering together with microscopic, XRD, and SEM–EDS characterization; it cannot be classified as a mechanochemical method merely because its subject is a powder [45].

Czochralski studies provide references for the melt, the Ir crucible, the atmosphere, SiO_2 volatilization, pulling and rotation, and annealing. The combined growth, morphology, and spectroscopic study of Y_2SiO_5 shows that structural and compositional identification is still required after crystal growth. The early Czochralski studies of rare-earth orthosilicates and the subsequent thesis can be used to standardize the reporting of process parameters, but there is at present no evidence of Czochralski growth of the target phase [4, 7, 31, 47].

Implications for experiments on the target phase. The contamination checks used during mechanochemical activation and the control of the melt, crucible, and volatilization in the Czochralski method can serve as references; at the same time, the bulk phase composition, the measured element ratios, inclusions, and the single-crystal structure must be verified. Ordinary mixing cannot be called mechanochemical activation, and the fact that $\text{Y}_2\text{SiO}_5/\text{Ln}_2\text{SiO}_5$ can be pulled by the Czochralski method does not imply that the target phase can be. All numerical values are limited to the original literature data listed in Table 2.

8 STRUCTURAL AND COMPOSITIONAL CHARACTERIZATION OF THE PRODUCTS

The four types of characterization listed on the right of Figure 3 address different questions and cannot substitute for one another: single-crystal or advanced diffraction methods confirm the structural model, PXRD/Rietveld analysis assesses the bulk phase composition, chemical analysis and microscopy check the measured element ratios and contamination, and high-temperature experiments delimit the stability range under a specific atmosphere and thermal history. The “Products and charac-

terization” column of Table 3 accordingly restricts the conclusions that each experimental record can support. When one type of data is missing, only the corresponding conclusion is held with caution; the gap cannot be filled by the other methods.

8.1 STRUCTURAL MODEL

Single-crystal X-ray diffraction anchors the identification of the target phase in a structural model of the measured crystal; the existing direct study identified $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ on exactly this basis [1]. In the related Zn-free Ba–Y silicates, single-crystal refinement revealed a non-stoichiometric modulated structure, and the combination of cRED, synchrotron PXRD, and neutron refinement confirmed an independent tetragonal Ba–Y phase region [18, 43]. Other single-crystal or synchrotron studies established the structures of their respective Ba–Y and Ba–Zn–Si compounds, but the conclusions apply only to those compounds [30, 51]. A single-crystal structure can prove that the measured crystal is consistent with the refined model, but it cannot by itself prove that the whole batch is phase-pure, nor can it equate the nominal starting composition with the measured site occupancy; local spectroscopy at most supplements the discrimination of polymorphs and provides no direct evidence of phase formation of the target phase [6].

8.2 BULK PHASE COMPOSITION

PXRD/Rietveld analysis addresses the bulk sample, focusing on phase fractions, unindexed diffraction peaks, and differences in polymorph under the same thermal history. The Ba–Y ceramic study detected secondary phases by powder diffraction, and the study of the independent phase region combined synchrotron powder data with other diffraction methods to confirm the phase composition [18, 44]; the Ba–Zn–Si studies likewise used X-ray, neutron, or synchrotron data to distinguish the bulk polymorph from the structural model [28, 51]. PXRD/Rietveld analysis can therefore support judgments about the bulk phase composition and impurity phases, but neither similar peak positions nor the appearance of the main phase automatically implies phase purity, and it cannot replace single-crystal determination of site occupancy. For studies that report only solid-state transitions or thermally induced changes, this review draws only on their characterization methods and does not treat the “optimized” in their titles as a criterion for the target phase [9, 23].

8.3 MEASURED COMPOSITION AND CONTAMINATION

The nominal ratio, the local spot compositions, and the overall sample composition need to be recorded separately. EDS, EPMA, or ICP can check compositional differences between grains, foreign phases, and material introduced by the processing media. The Zn-free Ba–Y ceramic study included both the secondary phases and the agate-derived SiO_2 contamination in its analysis [44]; the study of the independent Ba–Y phase region used EDS together with multiple diffraction methods to constrain the composition and structure [18]; and the related Ba/Sr–Zn–Si compounds were confirmed in their own compositional relationships by the combination of single-crystal structure and EDS [38]. These methods can only account for the elements detected within the measured spot or sampled volume; the homogeneity of a whole batch cannot be judged from a few spots, and the nominal pre-firing ratio cannot be rewritten directly as the post-firing chemical formula. Combined microscopic, thermal, and spectroscopic measurements can provide process references for doping or crystal growth, but they do not carry the compositional conclusions for the target phase [31, 35].

8.4 HIGH-TEMPERATURE STABILITY

HT-XRD, thermal analysis, dilatometry, and multi-atmosphere experiments examine how a material changes under a specific thermal history or atmosphere; they cannot replace room-temperature structural identification. The low-temperature and high-temperature phases of $\text{BaZn}_2\text{Si}_2\text{O}_7$ were distinguished by neutron and X-ray diffraction, showing that the in situ high-temperature phase cannot be equated with the product after cooling [28]; the related Zn/Co and Ba/Sr–Zn–Si/Ge studies further linked the substituted composition to thermal stability or phase boundaries [24, 36]. The tolerance of a related ceramic to a specific reducing atmosphere can serve only as an atmosphere reference and cannot prove that the target phase has the same stability [50]. High-temperature experiments can delimit the transition, decomposition, or atmosphere-tolerance range of a published system, but in situ

signals cannot by themselves prove that the cooled sample is phase-pure; where the corresponding data are lacking, the stability conclusion remains undetermined.

Together, the four types of characterization bound the scope of the conclusions: single-crystal or advanced diffraction confirms the structural model, powder diffraction assesses the bulk phase composition, chemical analysis verifies the measured element ratios and contamination, and high-temperature experiments delimit the thermal-history and atmosphere boundaries. Only when these results are matched to the synthesis conditions of the same batch can one discuss whether the target phase has been reproduced and how far the related processes can be extrapolated.

9 COMMON EXPERIMENTAL PROBLEMS AND SCOPE OF APPLICABILITY

The scope of applicability of a method can be judged only when failed or anomalous results are recorded together with the composition, the experimental conditions, and the product characterization. This section discusses phase competition, contamination and volatilization, atmosphere and polymorphism, missing information, and environmental, health, and safety (EHS) constraints; the content is confined to published conditions, and these boundaries are not rewritten as new experimental protocols.

Phase regions and phase competition. The systematic comparison of flux growth across the Ba–Y silicate family shows that phase-formation outcomes must be interpreted together with the composition and route variables; a single heat-treatment condition cannot be extracted in isolation as a general condition [42]. Subsequent studies, through re-examination of the non-stoichiometric structure, ceramic composition series, and multiple diffraction methods, revealed modulated structures, secondary phases, and competing phases or glass-forming regions [18, 43, 44]. These results delimit only the composition ranges that each study examined and cannot prove that no other phases exist in the uncovered regions. Mechanochemical studies indicate only that pre-activation may affect subsequent phase formation; a particular negative result cannot be interpreted as an intrinsic phase boundary of the target phase [41]. Consequently, the products, characterization, and synthesis conditions in Tables 2 and 3 must originate from the same publication, and impurity phases or glass cannot be interpreted in isolation from the composition and thermal history.

Contamination, volatilization, and flux boundaries. The Zn-free Ba–Y ceramic study linked the secondary phases and the agate-derived SiO₂ contamination to specific processing steps, showing that an apparent deviation from the nominal ratio should first prompt a check of the starting materials and the processing, and cannot automatically be taken as evidence of genuine solid solution [44]. The Czochralski study of the Y–Si–O system provides references for the melt, crucible, atmosphere, and pulling conditions, but the available material is insufficient to judge whether the target phase undergoes Zn or Si volatilization [47]. Lead- and fluorine-containing flux systems were once used for Zn–Si–O crystal growth, but these early conditions serve only for comparing slow cooling, separation, and the role of the flux, and cannot serve as a basis for material selection for the target phase [15]. Material introduced by the grinding media, mass loss in open systems, and flux residues or flux-driven phase selection should therefore be recorded separately; where the literature gives no pre- and post-firing compositions or residue analysis, this review does not assign a specific failure mechanism.

Atmosphere and polymorphism. The low-temperature monoclinic and high-temperature orthorhombic phases of BaZn₂Si₂O₇ were distinguished by X-ray and neutron diffraction, showing that the in situ high-temperature phase and the product after cooling are not interchangeable [28]. High-temperature diffraction further shows that Zn/Co substitution and the composition of the Ba/Sr–Zn–Si/Ge solid solutions are coupled to phase stability or polymorph boundaries; if any one of composition, temperature, or observation time point is missing, the stability range cannot be defined [24, 36]. The reduction-resistance report on a related Ba–Zn–Si ceramic shows only that that compound was examined for atmosphere tolerance under the published conditions and cannot be extrapolated to the reduction stability of the target phase [50]; other comparisons of thermal expansion and high-temperature diffraction likewise serve only as references for thermally induced changes [23].

Missing information and EHS. Some publications give only the synthesis method or the structural conclusion without also reporting the complete information on ratios, temperature–time, atmosphere,

crucible, and cooling [1, 2, 9, 20, 48]. This means that this review has not yet obtained the corresponding information; it does not mean that the original paper did not report it, nor that the step was absent from the experiment. Table 2 marks such entries with “—” and preserves this boundary. The less complete the information, the fewer specific conditions can be drawn upon. EHS boundaries are likewise described only according to the published conditions: Ba-containing powders, volatile species at high temperature, the Pb/F flux conditions in the early literature, crucible compatibility, and waste streams all require separate review, and the Pb/F routes in particular must not be rewritten from early examples into recommended protocols [15]. Any reuse must pass independent EHS, container-compatibility, and waste-segregation review; this review neither supplies specific operating parameters nor makes compliance presumptions about unreported ventilation, protective measures, equipment, or waste classification.

The four types of boundary above cannot be merged into a single unverified failure mechanism. Phase competition holds only within the published composition and thermal-history ranges; contamination or mass loss requires support from analysis of the media, of the pre- and post-firing compositions, or of residues; judgments about atmosphere and polymorphism must retain the observation temperature and the state after cooling; and missing information indicates only the limits of current knowledge. Accordingly, where the literature reports only “accompanied by” or gives only structural results, the text uses “accompanied by” or “consistent with” rather than “caused”. This preserves both negative results and records of success while avoiding conflation of unreported items, process anomalies, and genuine phase regions.

10 PRIORITY RESEARCH DIRECTIONS AND EXPERIMENTAL DESIGN

The following schemes are proposed jointly from the existing evidence and the model candidates; all are experimental hypotheses rather than verified recipes for the target phase. We recommend proceeding in the order “local phase diagram–site occupancy/oxygen stoichiometry–crystal growth–analogous substitution”; the phase-composition results of each stage determine whether the next stage is worth undertaking.

10.1 DIRECTION 1: A LOCAL PHASE DIAGRAM NEAR THE TARGET COMPOSITION

The first priority is to establish a fine-step composition grid around $\text{Ba}_5\text{Y}_{12}\text{ZnSi}_8\text{O}_{40}$. The base precursors are the model’s top-ranked and chemically conventional BaCO_3 , Y_2O_3 , ZnO , and SiO_2 , with the target stoichiometry corresponding to a molar ratio of 5:6:1:8. This direction is a model prediction awaiting experimental verification. We suggest first screening in stages over the temperature range of the Zn-free solid-state studies: an initial reaction at about 1000 °C for 12 h, followed by re-grinding and annealing at 1300 °C for 24 h, with repetition decided on the basis of the PXRD results; short-duration points at 1450–1600 °C are added only if obvious unreacted material remains and the mass loss and crucible compatibility are controllable. The temperatures and times are taken from the screening ranges of the neighboring Ba–Y solid-state studies [18, 44]; they serve only to organize the experimental matrix and do not imply that the target phase necessarily forms under these conditions.

The composition axes should include at least: (i) small deviations of the Ba/Y ratio around the target point; (ii) variation of the Zn content; and (iii) the Zn-free reference phase related to the oxygen stoichiometry. For every sample, the pre- and post-firing masses, the thermal history, and the cooling mode are recorded, and the phase composition and measured element ratios are given by quantitative PXRD/Rietveld analysis, SEM–EDS/EPMA, and ICP analysis where necessary. If glass or a multi-phase boundary appears, sampling should be densified in that region rather than retaining only the most ideal diffraction pattern.

10.2 DIRECTION 2: ZN–Y–OXYGEN-STOICHIOMETRY COUPLING

A testable charge-balanced series is

$$\text{Ba}_5\text{Y}_{13-x}\text{Zn}_x[\text{SiO}_4]_8\text{O}_{8.5-x/2}, \quad 0 \leq x \leq 1.25,$$

where $x = 0$ corresponds to the reported Zn-free tetragonal reference phase and $x = 1$ corresponds, in elemental stoichiometry, to the target composition. We suggest $x = 0, 0.25, 0.50, 0.75, 1.00, 1.25$ as the discrete points of the first round. This direction is a model prediction awaiting experimental

verification. The precursors remain BaCO_3 , Y_2O_3 , and SiO_2 , with the Zn content adjusted by ZnO ; the expression is only a charge-balance hypothesis and does not presuppose the true site occupancy of Zn or the distribution of oxygen vacancies. Assessment should examine simultaneously the continuity of the lattice parameters, the phase fractions, the measured Zn/Y ratio, and the site-occupancy refinement; where feasible, neutron diffraction or independent measurements of oxygen content and valence state can further distinguish Zn substitution, oxygen defects, and a two-phase mixture.

The scientific value of this series exceeds that of a single-point reproduction: a continuous change in the structural parameters supports limited solid solution; essentially unchanged lattice parameters with a changing two-phase ratio is closer to a two-phase region; and new diffraction peaks appearing only at $x \approx 1$ suggest a line compound. All three outcomes constrain the formation mechanism of the target phase.

10.3 DIRECTION 3: SOLID-STATE REACTION IN PARALLEL WITH HIGH-TEMPERATURE SOLUTION CRYSTAL GROWTH

The solid-state reaction route serves to locate the bulk phase region, and the high-temperature solution route serves to obtain crystals suitable for SCXRD analysis. This direction is a model prediction awaiting experimental verification. The high-temperature solution route can take the $\text{K}_2\text{CO}_3/\text{MoO}_3$ flux system already validated in the Ba–Y–Si–O system as its reference starting point, with the BaCO_3 , Y_2O_3 , ZnO , and SiO_2 precursors of Direction 1; the maximum temperature is set within 1100–1200 °C, held for about 3 h, followed by slow cooling at $1\text{--}2\text{ K h}^{-1}$ to about 900 °C, while the nutrient-to-flux ratio and the $\text{Y}_2\text{O}_3/\text{SiO}_2$ ratio are varied systematically [25, 42]. These values come from crystal growth in the Zn-free homologous series; they must be compared in parallel with the solid-state reaction route using the target starting composition and cannot be called a recipe for the target phase.

Each crystal-growth batch should report at least the liquid-phase composition, the crucible filling ratio, the maximum temperature, the dwell time, the cooling rate, the separation method, the crystal size, and the yield; the crystals are verified by SCXRD and EDS, and the residual mother liquor is checked by PXRD and compositional analysis. If only a few target single crystals are obtained while the composition of the mother liquor is far from the target composition, this should be interpreted as selective growth from the liquid phase rather than as a bulk phase region.

10.4 DIRECTION 4: Mg/Co ANALOGUES AND MODEL-PROPOSED PRECURSORS

This review invoked a local two-step inorganic retrosynthesis model to predict Top-5 precursor combinations for three elemental formulae. The model’s top-ranked combinations for the Zn, Mg, and Co targets all retain BaCO_3 , Y_2O_3 , and SiO_2 , with ZnO , MgO , and Co_3O_4 as the fourth starting material, respectively; the probabilities of the three top-ranked combinations within their respective candidate pools are 0.594, 0.522, and 0.298. This direction is a model prediction awaiting experimental verification. Table 4 converts the model output into minimal testable experiments. The model proposes precursor combinations solely on the basis of elemental stoichiometry and is not used to judge the silicate structural units; the text throughout retains the structural formula used in the target publication.

The remaining Zn routes given by the model include additional BaO , BaF_2 , or $\text{Ba}(\text{NO}_3)_2$, or non-standard database labels; their probabilities are all clearly lower than that of the top-ranked combination, and they should not be recommended alongside it. An additional Ba source would alter the local phase diagram; BaF_2 increases the risk of fluorine-containing volatiles and waste streams; and non-standard labels may further reflect normalization problems in the training data. MgCO_3 in the Mg routes can serve as a separate control for precursor form but should not be added together with MgO ; the several Co oxide precursors should not be regarded as three equivalent schemes.

10.5 ORGANIZING EXPERIMENTS FOR SCIENTIFIC DISCOVERY

Each round of experiments should be judged by the criterion of “maximizing distinguishable conclusions” rather than aiming merely at obtaining crystals. The most informative combination is: (i) a fine-step composition grid coupled with quantitative phase analysis to establish the local phase diagram; (ii) parallel solid-state reaction and flux-growth experiments at the same composition to

Table 4: Top-ranked precursor combinations from the two-step inorganic retrosynthesis model and their experimental uses. All combinations are model-ranked results (not validated by chemical routes); the process conditions are taken from the literature on neighboring systems and require re-checking by phase-diagram experiments.

Hypothetical target	Model top-ranked precursors	Candidate-pool probability	Suggested experiment and criteria
Zn target	$\text{BaCO}_3 + \text{Y}_2\text{O}_3 + \text{SiO}_2 + \text{ZnO}$	0.594	Base starting composition for the local phase diagram; solid-state screening first, then transfer of near-single-phase compositions to flux growth; bulk phase region and structural assignment determined by PXRD/Rietveld analysis and SCXRD, respectively
Mg analogue	$\text{BaCO}_3 + \text{Y}_2\text{O}_3 + \text{SiO}_2 + \text{MgO}$	0.522	Mg^{2+} as an isovalent size control; screening along the same composition/thermal-history matrix as for Zn; substitution on the same site discussed only if a new phase appears and the measured composition varies continuously
Co analogue	$\text{BaCO}_3 + \text{Y}_2\text{O}_3 + \text{SiO}_2 + \text{Co}_3\text{O}_4$	0.298	Air atmosphere as the first-round condition; measured composition and phase fractions determined; secondary phases not to be mistaken for structural substitution

distinguish the equilibrium phase region from selective nucleation in the liquid phase; and (iii) a coupled Zn-content and oxygen-stoichiometry series to test the site and defect hypotheses. All negative results are likewise preserved, because they determine the composition choices of the next round and constrain the boundary of “synthesizability”.

11 CONCLUSIONS

$\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ has reliable single-crystal structural evidence and evidence of synthesis by the high-temperature solution method, but no public evidence yet shows that its bulk phase-pure range has been independently reproduced [1]. At the present stage it is therefore inappropriate to give a “best recipe”; instead, the variables that must be controlled simultaneously and the characterization steps should be specified.

The neighboring systems provide three consistent insights. First, phase formation in the Ba–Y–Si–O system is sensitive to the composition coordinates, the MoO_3 content, and the cooling path; the Zn-free tetragonal phase additionally involves non-stoichiometry, competing phases, and a glass-forming region [18, 42, 43]. Second, secondary phases in ceramic samples alter the interpretation of the nominal composition, and a single-crystal assignment cannot substitute for bulk phase quantification [44]. Third, the polymorphism and phase stability of the Ba–Zn–Si–O system vary with thermal history and substituted composition, so the Zn source, the cooling path, and the observation temperature must be recorded jointly [24, 28, 50]. These insights can guide experimental design, but the numerical values of the related systems cannot be applied directly as conditions for the target phase.

The most informative work for the next stage is, in order: (i) establishing a local phase diagram near the target composition with BaCO_3 , Y_2O_3 , ZnO , and SiO_2 as the base starting materials; (ii) examining the $\text{Ba}_5\text{Y}_{13-x}\text{Zn}_x[\text{SiO}_4]_8\text{O}_{8.5-x/2}$ series to distinguish limited solid solution, a line compound, and a two-phase region; (iii) carrying out solid-state reaction and high-temperature solution crystal growth in parallel, determining the bulk phase region by PXRD/Rietveld analysis and the structural assignment by SCXRD; and (iv) testing the size difference by Mg/Co analogy only after the target phase has been independently reproduced. The oxide or carbonate combinations recommended by the retrosynthesis model can serve as a starting point for precursors but cannot replace phase-diagram, thermal-history, and structural verification.

The experiment most worth starting immediately is a fine-step local phase diagram near the target composition: using BaCO₃, Y₂O₃, ZnO, and SiO₂ as precursors, performing staged calcination and annealing within the temperature range of the neighboring systems, and subjecting each composition point to quantitative PXRD/Rietveld analysis and EDS or EPMA analysis, with priority given to establishing whether the target phase has a reproducible bulk phase region; the near-single-phase compositions are then transferred to high-temperature solution crystal growth, and the structure is re-checked by SCXRD. Such experiments update the hypotheses whether they succeed or fail. A fine-step composition grid, the measured element ratios and oxygen stoichiometry, cross-validation between single-crystal and powder diffraction, and fully preserved negative results together form an iterable experimental basis and clarify the following questions: whether Zn enters the Zn-free tetragonal silicate series, whether the target phase has a reproducible stability region, and whether the crystal-growth conditions and the bulk phase-formation conditions belong to the same phase-equilibrium region.

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